

Mohammed Alaa

AI Developer | AI Infrastructure Engineer | Data Scientist

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PROFESSIONAL SUMMARY

AI Developer and Infrastructure Engineer with experience building, testing, and deploying production-ready AI solutions. Skilled in Large Language Models (LLMs), Prompt Engineering, RAG systems, API integrations, and AI-driven automation. Strong foundation in Computer Science, mathematics, and high-performance computing, with hands-on experience developing research-oriented C++ projects in quantum simulation and photonic computing. Experienced in remote and startup environments, with a focus on problem-solving, scalability, and cross-functional collaboration.

PROJECTS

Phoenix-AI (Production SDK) ([Docs](#) | [GitHub](#))

Modular backend infrastructure SDK for AI-powered workflows and agentic Python services, built for scale-up and startup environments.

- LLM and API Integration: Interface-first design (BaseLLM, BaseVectorDB, BaseCache) enabling plug-and-play swapping of AI providers and automation platforms.
- RAG Pipeline: End-to-end retrieval-augmented generation (RAG) handling PDF/DOCX/XLSX documents, SQL databases, APIs, and web scraping.
- Vector Databases: Native support for ChromaDB and Qdrant with embeddings via sentence-transformers (all-MiniLM-L6-v2).
- Prompt Engineering layer with dependency injection registry for clean, testable LLM interaction patterns.
- Sensorium Hardware SDK: AI integration with real hardware (Serial, MQTT, GPIO, Cameras) for embodied agentic systems.

Ashborn -- AI Agent ([GitHub](#))

Fully autonomous AI Developer agent built for remote, contract-style code generation tasks -- transforms natural-language intent into production-ready multi-file projects.

- Implements a multi-phase cognition loop: strategic thinking, workflow design, execution, and self-correction -- no human intervention required.
- Ships as both a standalone IDE (VSCodium-powered) and terminal interface; demonstrates systems thinking at the AI workflow level.

System Stability Guardian -- AI Agent ([GitHub](#))

designed exclusively to ensure operational calmness and system integrity. Unlike general-purpose agents, it is a highly specialized diagnostic sentinel operating on a **Hybrid Cognitive Architecture** that prioritizes precision, deterministic safety, and real-time anomaly detection over open-ended generation.

- **Continuous Health Monitoring:** Scanning hardware metrics, agent heartbeats, and process states.
- **Deterministic Escalation:** Bypassing LLM hallucinations by using strict, rule-based logic to decide when a system warning should become a critical escalation.
- **Adaptive Context Logging:** Injecting real-time health data (stability_reports, anomalies, escalation_state) directly into the agent network's backbone (giyu.context.json), ensuring all other agents are aware of the system's condition.

Shinobu Intelligent Search -- AI Agent ([GitHub](#))

understanding your intent in natural language, planning the required actions, and delivering results in a clear, friendly, and actionable format.

a tiered research engine that adapts to the complexity of your request:

- **Fast Search:** Instant redirection. Best for "Open YouTube" or "Go to Twitter".
- **Mid Search:** Result aggregation. Scrapes search engines to provide a structured overview of links and snippets.
- **Deep Search:** Comprehensive analysis. Scrapes multiple high-authority sources, extracts full-text content, and uses LLM synthesis to provide a detailed research report.
- **Auto Mode:** The default setting where Shinobu's IntentInterpreter decides the optimal depth based on your natural language prompt.

Phoenix-LLM ([GitHub](#))

From-scratch decoder-only Transformer LLM built on a custom framework -- demonstrates deep understanding of LLMs (Large Language Models) at the architecture level.

- Implements Grouped Query Attention (GQA), SwiGLU FFN, RMSNorm Pre-LN, and weight tying for optimised inference and training stability.
- Fully modular: swappable attention, FFN, normalisation, and positional encoding components for rapid experimentation.
- Current Progress: Phase 3 -- model architecture complete; distributed training loop and streaming inference API in development.

WORK EXPERIENCE

Software Developer -- Digithub, Egypt

Sep 2025 - Apr 2026

- Designed and deployed AI integration points within 3 production Django applications, applying LLM-powered automation to reduce manual processing time.
- Developed and maintained 80+ RESTful API endpoints using Django and DRF, with full API integration across Admin Dashboards and User Web Applications.
- Implemented workflow design for async task processing using Celery and Redis, improving business efficiency across order and resource management systems.
- Built RBAC permission systems supporting 5+ distinct user roles, demonstrating high attention to detail in complex access-control logic.
- Collaborated with stakeholders across 3 parallel production systems within agile sprint cycles, demonstrating adaptability in a fast-paced environment:
 1. Real Estate Management System
 2. Learning Management System (LMS)
 3. Car Rental Management System

EDUCATION

Bachelor of Science in Computer Science -- South Valley University, Egypt

Oct 2022 - Jun 2026

SKILLS

AI & Machine Learning: LLM Integration (OpenAI, Claude), Prompt Engineering, RAG Pipelines, AI-Powered Workflows, AI Solution Deployment, Embeddings & Vector Similarity Search, sentence-transformers, TensorFlow, PyTorch, scikit-learn, pandas, numpy.

Vector Databases & Knowledge Stores: ChromaDB, Qdrant, PostgreSQL, MySQL, SQLite, Database Design.

Backend Development: Python, C++, Django, FastAPI, Django REST Framework, Celery & Redis, Async Task Queues.

Frontend Development: HTML, CSS, JavaScript, Bootstrap.

Applied & Emerging Computing: Quantum Computing, Optical Neural Networks (ONNs).

Infrastructure & Practices: Docker, Git, Clean Architecture, SOLID Principles, Agile.

Soft Skills: Cross-functional Collaboration, Problem Solving, Technical Communication.

Languages

- Arabic (Native)
- English

CERTIFICATIONS

HCIA - AI V3.5 -- Huawei

Feb 2025